rpg_demo

November 12, 2026

GitHub: https://github.com/OleKrGaustad/INF201_Deliverable3

This notebook demonstrates all classes created:

- Dice
- Weapon
- Character (base class)
- Warrior (defensive specialist)
- Mage (resource-based attacker)
- Battle (combat manager)

Structure 1. Imports 2. Presentation of each class (attributes + methods) 3. Simulated battles between different character types

1. Imports

[2]:
```
from dice import Dice
from items import Weapon
from character import Character, Warrior, Mage
from battle import Battle
import inspect
import venv
```

2. Presentation of each class

For every class we show: - Short description - Public attributes (instance variables) - Public methods and properties - A small live example

0.0.1 2.1 Dice class

Purpose: Introduce randomness into damage calculations.

Attributes: - `sides` (int) – number of sides on the dice (2)

Methods: - `roll()` -> returns a random int between 1 and `sides` - `__call__()` -> allows `d20()` instead of `d20.roll()` - `__repr__()` -> nice string representation

[3]:
```
print("=== Dice class overview ===")
print("Public members:")
for name, member in inspect.getmembers(Dice):
    if name in ("__init__", "__repr__", "__call__", "roll") or (not name.
 ↪startswith("_")):
```

```
        print(f"  • {name}: {type(member).__name__}")

print("\n--- Live example ---")
d6 = Dice()               # default 6-sided
d20 = Dice(sides=20)
d4 = Dice(4)

print(d6)
print(d20)
print(d4)
print()
print("d6.roll() ->", d6.roll())
print("d20()     ->", d20())      # uses __call__
print("d4()      ->", d4())
print()
print("Attributes of d20:", vars(d20))
```

```
=== Dice class overview ===
Public members:
  • __call__: function
  • __init__: function
  • __repr__: function
  • roll: function

--- Live example ---
Dice(sides=6)
Dice(sides=20)
Dice(sides=4)

d6.roll() -> 4
d20()     -> 19
d4()      -> 1

Attributes of d20: {'sides': 20}
```

0.0.2 2.2 Weapon class

Purpose: Represent a weapon that deals variable damage using a Dice.

Attributes: - `name` (str) - `damage_dice` (Dice)

Methods: - `roll_damage()` -> rolls the weapon's dice and returns the damage - `__repr__()` / `__str__()`

[4]:
```
print("=== Weapon class overview ===")
print("Public members:")
for name, member in inspect.getmembers(Weapon):
    if name in ("__init__", "__repr__", "__str__", "roll_damage") or (not name.
 ↪startswith("_")):
```

```
        print(f"  • {name}: {type(member).__name__}")

print("\n--- Live example ---")
longsword = Weapon("Longsword", Dice(8))
dagger = Weapon("Dagger", Dice(4))
fire_staff = Weapon("Fire Staff", Dice(10))

print(longsword)
print(dagger)
print(fire_staff)
print()
print("Damage rolls:")
print("  Longsword :", longsword.roll_damage())
print("  Dagger    :", dagger.roll_damage())
print("  Fire Staff:", fire_staff.roll_damage())
print()
print("Attributes of longsword:", vars(longsword))
```

```
=== Weapon class overview ===
Public members:
  • __init__: function
  • __repr__: function
  • __str__: function
  • roll_damage: function

--- Live example ---
Longsword (Dice(sides=8))
Dagger (Dice(sides=4))
Fire Staff (Dice(sides=10))

Damage rolls:
  Longsword : 2
  Dagger    : 2
  Fire Staff: 6

Attributes of longsword: {'name': 'Longsword', 'damage_dice': Dice(sides=8)}
```

0.0.3 2.3 Character class (base class)

Purpose: Base class for all characters. Handles HP, basic attacks and encapsulation.

Attributes: - `name` (str) - `_hp` (private) – current hit points - `hp_max` (int) - `damage` (int) – base damage used when no weapon is equipped - `weapon` (Weapon | None)

Properties (read-only): - `hp` -> current hit points - `is_alive` -> `True` if `_hp > 0`

Methods: - `take_damage(amount)` – reduces own HP (never below 0) - `heal(amount)` – restores HP (never above `hp_max`) - `attack(target)` – deals damage to another character (uses weapon if equipped) - `introduce()` - `__repr__()` / `__str__()`

[5]:
```
print("=== Character class overview ===")
print("Public members:")
for name, member in inspect.getmembers(Character):
    if name in ("__init__", "__repr__", "__str__", "attack", "heal",
                "take_damage", "introduce", "hp", "is_alive") or (not name.
 ↪startswith("_")):
        print(f"  • {name}: {type(member).__name__}")

print("\n--- Live example ---")
hero = Character("Aria", hp_max=100, damage=12, weapon=longsword)
goblin = Character("Goblin", hp_max=40, damage=6)   # no weapon

print(hero)
print(goblin)
print()
print("hero.hp        ->", hero.hp)
print("hero.is_alive  ->", hero.is_alive)
print("hero.weapon    ->", hero.weapon)
print()
print("--- Hero attacks Goblin ---")
hero.attack(goblin)
print(goblin)
print()
print("--- Goblin takes heavy damage ---")
goblin.take_damage(50)
print(goblin)
print("goblin.is_alive ->", goblin.is_alive)
```

```
=== Character class overview ===
Public members:
  • __init__: function
  • __repr__: function
  • __str__: function
  • attack: function
  • heal: function
  • hp: property
  • introduce: function
  • is_alive: property
  • take_damage: function

--- Live example ---
Character: Aria (100/100 HP, 12 base dmg, weapon=Longsword) [alive]
Character: Goblin (40/40 HP, 6 base dmg) [alive]

hero.hp        -> 100
hero.is_alive  -> True
hero.weapon    -> Longsword (Dice(sides=8))
```

```
--- Hero attacks Goblin ---
Character: Goblin (39/40 HP, 6 base dmg) [alive]

--- Goblin takes heavy damage ---
Character: Goblin (0/40 HP, 6 base dmg) [defeated]
goblin.is_alive -> False
```

0.0.4 2.4 Warrior class (subclass of Character)

Purpose: Defensive specialist.

New attribute: - `armor` (int, default = 5)

Overridden method: - `take_damage(amount)` – incoming damage is reduced by `armor` before being applied

Inherits everything else from `Character` (`hp`, `is_alive`, `attack`, `heal`, etc.).

[6]:
```
print("=== Warrior class overview ===")
print("Public members:")
for name, member in inspect.getmembers(Warrior):
    if name in ("__init__", "__repr__", "__str__", "attack", "heal",
                "take_damage", "introduce", "hp", "is_alive") or (not name.
 ↪startswith("_")):
        print(f"  • {name}: {type(member).__name__}")

print("\n--- Live example ---")
warrior = Warrior("Thorgar", hp_max=120, damage=14, weapon=longsword, armor=7)
print(warrior)
print()
print("warrior.armor ->", warrior.armor)
print()
print("Warrior takes 20 raw damage (armor reduces it by 7):")
print("HP before:", warrior.hp)
warrior.take_damage(20)
print("HP after :", warrior.hp)   # 120 - (20-7) = 107
```

```
=== Warrior class overview ===
Public members:
  • __init__: function
  • __repr__: function
  • __str__: function
  • attack: function
  • heal: function
  • hp: property
  • introduce: function
  • is_alive: property
  • take_damage: function

--- Live example ---
```

```
Warrior: Thorgar (120/120 HP, 14 base dmg, armor=7, weapon=Longsword) [alive]

warrior.armor -> 7

Warrior takes 20 raw damage (armor reduces it by 7):
HP before: 120
HP after : 107
```

0.0.5 2.5 Mage class (subclass of Character)

Purpose: Resource-based magical attacker.

New attributes: - `mana` (int) - `mana_max` (int, default = 30)

Overridden method: - `attack(target)`: - If `mana >= 10` -> spend 10 mana and deal **double damage** (magic attack, ignores weapon) - If `mana < 10` -> normal attack (uses weapon if equipped) **and restore 5 mana**

[7]:
```
print("=== Mage class overview ===")
print("Public members:")
for name, member in inspect.getmembers(Mage):
    if name in ("__init__", "__repr__", "__str__", "attack", "heal",
                "take_damage", "introduce", "hp", "is_alive") or (not name.
 ↪startswith("_")):
        print(f"  • {name}: {type(member).__name__}")

print("\n--- Live example ---")
mage = Mage("Elara", hp_max=80, damage=10, weapon=fire_staff, mana_max=30)
print(mage)
print()
print("mage.mana     ->", mage.mana)
print("mage.mana_max ->", mage.mana_max)
print()

dummy = Character("Training Dummy", hp_max=200, damage=1)

print("--- Mage attacks (shows mana system) ---")
for i in range(4):
    print(f"Attack {i+1}: mana before = {mage.mana}")
    mage.attack(dummy)
    print(f"           mana after  = {mage.mana}, Dummy HP = {dummy.hp}")
    print()
```

```
=== Mage class overview ===
Public members:
  • __init__: function
  • __repr__: function
  • __str__: function
  • attack: function
```

- heal: function
- hp: property
- introduce: function
- is_alive: property
- take_damage: function

```
--- Live example ---
Mage: Elara (80/80 HP, 10 base dmg, mana=30/30, weapon=Fire Staff) [alive]

mage.mana     -> 30
mage.mana_max -> 30

--- Mage attacks (shows mana system) ---
Attack 1: mana before = 30
          mana after  = 20, Dummy HP = 180

Attack 2: mana before = 20
          mana after  = 10, Dummy HP = 160

Attack 3: mana before = 10
          mana after  = 0, Dummy HP = 140

Attack 4: mana before = 0
          mana after  = 5, Dummy HP = 136
```

0.0.6 2.6 Battle class

Purpose: Manage turn-based combat between two characters.

Attributes: - `hero` (Character) - `enemy` (Character) - `round_number` (int)

Methods: - `play_round()` – hero attacks, then enemy attacks (if still alive) - `simulate_battle(verbose=True)` – runs rounds until one character is defeated and returns the winner

[8]:
```
print("=== Battle class overview ===")
print("Public members:")
for name, member in inspect.getmembers(Battle):
    if name in ("__init__", "__repr__", "play_round", "simulate_battle") or␣
 ↪(not name.startswith("_")):
        print(f"  • {name}: {type(member).__name__}")

print("\n--- Live example (short fight) ---")
fighter = Character("Rogan", hp_max=60, damage=15, weapon=Weapon("Axe",␣
 ↪Dice(8)))
bandit = Character("Bandit", hp_max=45, damage=8)

battle = Battle(hero=fighter, enemy=bandit)
```

```python
print(battle)
print()
winner = battle.simulate_battle(verbose=True)
print(f"\nWinner: {winner.name}")
```

```
=== Battle class overview ===
Public members:
  • __init__: function
  • __repr__: function
  • play_round: function
  • simulate_battle: function

--- Live example (short fight) ---
<Battle hero='Rogan' enemy='Bandit' round=0>

============================================================
BATTLE START: Rogan vs Bandit
============================================================
  Character: Rogan (60/60 HP, 15 base dmg, weapon=Axe) [alive]
  Character: Bandit (45/45 HP, 8 base dmg) [alive]

--- Round 1 ---
  Character: Rogan (52/60 HP, 15 base dmg, weapon=Axe) [alive]
  Character: Bandit (44/45 HP, 8 base dmg) [alive]

--- Round 2 ---
  Character: Rogan (44/60 HP, 15 base dmg, weapon=Axe) [alive]
  Character: Bandit (39/45 HP, 8 base dmg) [alive]

--- Round 3 ---
  Character: Rogan (36/60 HP, 15 base dmg, weapon=Axe) [alive]
  Character: Bandit (38/45 HP, 8 base dmg) [alive]

--- Round 4 ---
  Character: Rogan (28/60 HP, 15 base dmg, weapon=Axe) [alive]
  Character: Bandit (30/45 HP, 8 base dmg) [alive]

--- Round 5 ---
  Character: Rogan (20/60 HP, 15 base dmg, weapon=Axe) [alive]
  Character: Bandit (23/45 HP, 8 base dmg) [alive]

--- Round 6 ---
  Character: Rogan (12/60 HP, 15 base dmg, weapon=Axe) [alive]
  Character: Bandit (19/45 HP, 8 base dmg) [alive]

--- Round 7 ---
  Character: Rogan (4/60 HP, 15 base dmg, weapon=Axe) [alive]
  Character: Bandit (17/45 HP, 8 base dmg) [alive]
```

```
--- Round 8 ---
  Character: Rogan (0/60 HP, 15 base dmg, weapon=Axe) [defeated]
  Character: Bandit (15/45 HP, 8 base dmg) [alive]

============================================================
BATTLE OVER - Winner: Bandit
============================================================

Winner: Bandit
```

0.1 3. Simulated Battles

0.1.1 3.1 Warrior vs Mage (both with weapons)

[9]:
```python
hero = Warrior(
    "Thorgar",
    hp_max=130,
    damage=13,
    weapon=Weapon("Longsword", Dice(8)),
    armor=6
)

enemy = Mage(
    "Dark Elara",
    hp_max=95,
    damage=11,
    weapon=Weapon("Arcane Staff", Dice(6)),
    mana_max=40
)

battle1 = Battle(hero=hero, enemy=enemy)
winner1 = battle1.simulate_battle(verbose=True)

print(f"\nWinner of Battle 1: {winner1.name}")
```

```
============================================================
BATTLE START: Thorgar vs Dark Elara
============================================================
  Warrior: Thorgar (130/130 HP, 13 base dmg, armor=6, weapon=Longsword) [alive]
  Mage: Dark Elara (95/95 HP, 11 base dmg, mana=40/40, weapon=Arcane Staff)
[alive]

--- Round 1 ---
  Warrior: Thorgar (114/130 HP, 13 base dmg, armor=6, weapon=Longsword) [alive]
  Mage: Dark Elara (92/95 HP, 11 base dmg, mana=30/40, weapon=Arcane Staff)
[alive]
```

```
--- Round 2 ---
  Warrior: Thorgar (98/130 HP, 13 base dmg, armor=6, weapon=Longsword) [alive]
  Mage: Dark Elara (88/95 HP, 11 base dmg, mana=20/40, weapon=Arcane Staff)
[alive]

--- Round 3 ---
  Warrior: Thorgar (82/130 HP, 13 base dmg, armor=6, weapon=Longsword) [alive]
  Mage: Dark Elara (83/95 HP, 11 base dmg, mana=10/40, weapon=Arcane Staff)
[alive]

--- Round 4 ---
  Warrior: Thorgar (66/130 HP, 13 base dmg, armor=6, weapon=Longsword) [alive]
  Mage: Dark Elara (78/95 HP, 11 base dmg, mana=0/40, weapon=Arcane Staff)
[alive]

--- Round 5 ---
  Warrior: Thorgar (66/130 HP, 13 base dmg, armor=6, weapon=Longsword) [alive]
  Mage: Dark Elara (70/95 HP, 11 base dmg, mana=5/40, weapon=Arcane Staff)
[alive]

--- Round 6 ---
  Warrior: Thorgar (66/130 HP, 13 base dmg, armor=6, weapon=Longsword) [alive]
  Mage: Dark Elara (62/95 HP, 11 base dmg, mana=10/40, weapon=Arcane Staff)
[alive]

--- Round 7 ---
  Warrior: Thorgar (50/130 HP, 13 base dmg, armor=6, weapon=Longsword) [alive]
  Mage: Dark Elara (59/95 HP, 11 base dmg, mana=0/40, weapon=Arcane Staff)
[alive]

--- Round 8 ---
  Warrior: Thorgar (50/130 HP, 13 base dmg, armor=6, weapon=Longsword) [alive]
  Mage: Dark Elara (55/95 HP, 11 base dmg, mana=5/40, weapon=Arcane Staff)
[alive]

--- Round 9 ---
  Warrior: Thorgar (50/130 HP, 13 base dmg, armor=6, weapon=Longsword) [alive]
  Mage: Dark Elara (51/95 HP, 11 base dmg, mana=10/40, weapon=Arcane Staff)
[alive]

--- Round 10 ---
  Warrior: Thorgar (34/130 HP, 13 base dmg, armor=6, weapon=Longsword) [alive]
  Mage: Dark Elara (44/95 HP, 11 base dmg, mana=0/40, weapon=Arcane Staff)
[alive]

--- Round 11 ---
  Warrior: Thorgar (34/130 HP, 13 base dmg, armor=6, weapon=Longsword) [alive]
  Mage: Dark Elara (38/95 HP, 11 base dmg, mana=5/40, weapon=Arcane Staff)
```

```
[alive]

--- Round 12 ---
  Warrior: Thorgar (34/130 HP, 13 base dmg, armor=6, weapon=Longsword) [alive]
  Mage: Dark Elara (30/95 HP, 11 base dmg, mana=10/40, weapon=Arcane Staff)
[alive]

--- Round 13 ---
  Warrior: Thorgar (18/130 HP, 13 base dmg, armor=6, weapon=Longsword) [alive]
  Mage: Dark Elara (22/95 HP, 11 base dmg, mana=0/40, weapon=Arcane Staff)
[alive]

--- Round 14 ---
  Warrior: Thorgar (18/130 HP, 13 base dmg, armor=6, weapon=Longsword) [alive]
  Mage: Dark Elara (16/95 HP, 11 base dmg, mana=5/40, weapon=Arcane Staff)
[alive]

--- Round 15 ---
  Warrior: Thorgar (18/130 HP, 13 base dmg, armor=6, weapon=Longsword) [alive]
  Mage: Dark Elara (12/95 HP, 11 base dmg, mana=10/40, weapon=Arcane Staff)
[alive]

--- Round 16 ---
  Warrior: Thorgar (2/130 HP, 13 base dmg, armor=6, weapon=Longsword) [alive]
  Mage: Dark Elara (4/95 HP, 11 base dmg, mana=0/40, weapon=Arcane Staff)
[alive]

--- Round 17 ---
  Warrior: Thorgar (2/130 HP, 13 base dmg, armor=6, weapon=Longsword) [alive]
  Mage: Dark Elara (2/95 HP, 11 base dmg, mana=5/40, weapon=Arcane Staff)
[alive]

--- Round 18 ---
  Warrior: Thorgar (2/130 HP, 13 base dmg, armor=6, weapon=Longsword) [alive]
  Mage: Dark Elara (0/95 HP, 11 base dmg, mana=5/40, weapon=Arcane Staff)
[defeated]

==========================================================
BATTLE OVER - Winner: Thorgar
==========================================================

Winner of Battle 1: Thorgar
```

0.1.2 3.2 Two base Characters (one without weapon)

[10]:
```
fighter = Character("Rogan", hp_max=100, damage=15, weapon=Weapon("Battle Axe",␣
 ↪Dice(10)))
bandit = Character("Bandit", hp_max=70, damage=9)  # no weapon

battle2 = Battle(hero=fighter, enemy=bandit)
winner2 = battle2.simulate_battle(verbose=True)

print(f"\nWinner of Battle 2: {winner2.name}")
```

```
============================================================
BATTLE START: Rogan vs Bandit
============================================================
  Character: Rogan (100/100 HP, 15 base dmg, weapon=Battle Axe) [alive]
  Character: Bandit (70/70 HP, 9 base dmg) [alive]

--- Round 1 ---
  Character: Rogan (91/100 HP, 15 base dmg, weapon=Battle Axe) [alive]
  Character: Bandit (63/70 HP, 9 base dmg) [alive]

--- Round 2 ---
  Character: Rogan (82/100 HP, 15 base dmg, weapon=Battle Axe) [alive]
  Character: Bandit (57/70 HP, 9 base dmg) [alive]

--- Round 3 ---
  Character: Rogan (73/100 HP, 15 base dmg, weapon=Battle Axe) [alive]
  Character: Bandit (49/70 HP, 9 base dmg) [alive]

--- Round 4 ---
  Character: Rogan (64/100 HP, 15 base dmg, weapon=Battle Axe) [alive]
  Character: Bandit (42/70 HP, 9 base dmg) [alive]

--- Round 5 ---
  Character: Rogan (55/100 HP, 15 base dmg, weapon=Battle Axe) [alive]
  Character: Bandit (38/70 HP, 9 base dmg) [alive]

--- Round 6 ---
  Character: Rogan (46/100 HP, 15 base dmg, weapon=Battle Axe) [alive]
  Character: Bandit (36/70 HP, 9 base dmg) [alive]

--- Round 7 ---
  Character: Rogan (37/100 HP, 15 base dmg, weapon=Battle Axe) [alive]
  Character: Bandit (32/70 HP, 9 base dmg) [alive]

--- Round 8 ---
  Character: Rogan (28/100 HP, 15 base dmg, weapon=Battle Axe) [alive]
  Character: Bandit (30/70 HP, 9 base dmg) [alive]
```

```
--- Round 9 ---
  Character: Rogan (19/100 HP, 15 base dmg, weapon=Battle Axe) [alive]
  Character: Bandit (22/70 HP, 9 base dmg) [alive]

--- Round 10 ---
  Character: Rogan (10/100 HP, 15 base dmg, weapon=Battle Axe) [alive]
  Character: Bandit (14/70 HP, 9 base dmg) [alive]

--- Round 11 ---
  Character: Rogan (1/100 HP, 15 base dmg, weapon=Battle Axe) [alive]
  Character: Bandit (8/70 HP, 9 base dmg) [alive]

--- Round 12 ---
  Character: Rogan (0/100 HP, 15 base dmg, weapon=Battle Axe) [defeated]
  Character: Bandit (1/70 HP, 9 base dmg) [alive]

==========================================================
BATTLE OVER - Winner: Bandit
==========================================================

Winner of Battle 2: Bandit
```

0.1.3 3.3 Mage with low starting mana (demonstrates fallback attack)

[11]:
```
mage2 = Mage("Lyra", hp_max=85, damage=12, weapon=Weapon("Crystal Wand",␣
 ↪Dice(4)), mana_max=25)
mage2.mana = 5   # force low mana so we see the fallback + mana restore

warrior2 = Warrior("Borin", hp_max=110, damage=11, weapon=Weapon("Warhammer",␣
 ↪Dice(8)), armor=5)

print("Starting state:")
print(f"  {mage2}")
print(f"  {warrior2}")
print()

battle3 = Battle(hero=mage2, enemy=warrior2)
winner3 = battle3.simulate_battle(verbose=True)

print(f"\nWinner of Battle 3: {winner3.name}")
```

```
Starting state:
  Mage: Lyra (85/85 HP, 12 base dmg, mana=5/25, weapon=Crystal Wand) [alive]
  Warrior: Borin (110/110 HP, 11 base dmg, armor=5, weapon=Warhammer) [alive]

==========================================================
BATTLE START: Lyra vs Borin
```

```
======================================================
  Mage: Lyra (85/85 HP, 12 base dmg, mana=5/25, weapon=Crystal Wand) [alive]
  Warrior: Borin (110/110 HP, 11 base dmg, armor=5, weapon=Warhammer) [alive]

--- Round 1 ---
  Mage: Lyra (83/85 HP, 12 base dmg, mana=10/25, weapon=Crystal Wand) [alive]
  Warrior: Borin (110/110 HP, 11 base dmg, armor=5, weapon=Warhammer) [alive]

--- Round 2 ---
  Mage: Lyra (76/85 HP, 12 base dmg, mana=0/25, weapon=Crystal Wand) [alive]
  Warrior: Borin (91/110 HP, 11 base dmg, armor=5, weapon=Warhammer) [alive]

--- Round 3 ---
  Mage: Lyra (70/85 HP, 12 base dmg, mana=5/25, weapon=Crystal Wand) [alive]
  Warrior: Borin (91/110 HP, 11 base dmg, armor=5, weapon=Warhammer) [alive]

--- Round 4 ---
  Mage: Lyra (68/85 HP, 12 base dmg, mana=10/25, weapon=Crystal Wand) [alive]
  Warrior: Borin (91/110 HP, 11 base dmg, armor=5, weapon=Warhammer) [alive]

--- Round 5 ---
  Mage: Lyra (65/85 HP, 12 base dmg, mana=0/25, weapon=Crystal Wand) [alive]
  Warrior: Borin (72/110 HP, 11 base dmg, armor=5, weapon=Warhammer) [alive]

--- Round 6 ---
  Mage: Lyra (62/85 HP, 12 base dmg, mana=5/25, weapon=Crystal Wand) [alive]
  Warrior: Borin (72/110 HP, 11 base dmg, armor=5, weapon=Warhammer) [alive]

--- Round 7 ---
  Mage: Lyra (55/85 HP, 12 base dmg, mana=10/25, weapon=Crystal Wand) [alive]
  Warrior: Borin (72/110 HP, 11 base dmg, armor=5, weapon=Warhammer) [alive]

--- Round 8 ---
  Mage: Lyra (50/85 HP, 12 base dmg, mana=0/25, weapon=Crystal Wand) [alive]
  Warrior: Borin (53/110 HP, 11 base dmg, armor=5, weapon=Warhammer) [alive]

--- Round 9 ---
  Mage: Lyra (44/85 HP, 12 base dmg, mana=5/25, weapon=Crystal Wand) [alive]
  Warrior: Borin (53/110 HP, 11 base dmg, armor=5, weapon=Warhammer) [alive]

--- Round 10 ---
  Mage: Lyra (40/85 HP, 12 base dmg, mana=10/25, weapon=Crystal Wand) [alive]
  Warrior: Borin (53/110 HP, 11 base dmg, armor=5, weapon=Warhammer) [alive]

--- Round 11 ---
  Mage: Lyra (39/85 HP, 12 base dmg, mana=0/25, weapon=Crystal Wand) [alive]
  Warrior: Borin (34/110 HP, 11 base dmg, armor=5, weapon=Warhammer) [alive]
```

```
--- Round 12 ---
  Mage: Lyra (37/85 HP, 12 base dmg, mana=5/25, weapon=Crystal Wand) [alive]
  Warrior: Borin (34/110 HP, 11 base dmg, armor=5, weapon=Warhammer) [alive]

--- Round 13 ---
  Mage: Lyra (29/85 HP, 12 base dmg, mana=10/25, weapon=Crystal Wand) [alive]
  Warrior: Borin (34/110 HP, 11 base dmg, armor=5, weapon=Warhammer) [alive]

--- Round 14 ---
  Mage: Lyra (22/85 HP, 12 base dmg, mana=0/25, weapon=Crystal Wand) [alive]
  Warrior: Borin (15/110 HP, 11 base dmg, armor=5, weapon=Warhammer) [alive]

--- Round 15 ---
  Mage: Lyra (15/85 HP, 12 base dmg, mana=5/25, weapon=Crystal Wand) [alive]
  Warrior: Borin (15/110 HP, 11 base dmg, armor=5, weapon=Warhammer) [alive]

--- Round 16 ---
  Mage: Lyra (14/85 HP, 12 base dmg, mana=10/25, weapon=Crystal Wand) [alive]
  Warrior: Borin (15/110 HP, 11 base dmg, armor=5, weapon=Warhammer) [alive]

--- Round 17 ---
  Mage: Lyra (14/85 HP, 12 base dmg, mana=0/25, weapon=Crystal Wand) [alive]
  Warrior: Borin (0/110 HP, 11 base dmg, armor=5, weapon=Warhammer) [defeated]

======================================================
BATTLE OVER - Winner: Lyra
======================================================

Winner of Battle 3: Lyra
```